

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

1. Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks. There are two existing strategies for applying pre-trained language representations to downstream tasks: feature-based and fine-tuning. We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches, because standard language models are unidirectional, limiting the choice of architectures during pre-training.

In this paper we improve the fine-tuning based approaches by proposing BERT: Bidirectional Encoder Representations from Transformers. BERT alleviates the previously mentioned unidirectionality constraint by using a masked language model (MLM) pre-training objective, inspired by the Cloze task.

3. BERT

3.1 Pre-training BERT. Unlike Peters et al. and Radford et al., we do not use traditional left-to-right or right-to-left language models to pre-train BERT. Instead, we pre-train BERT using two unsupervised tasks.

Task #1: Masked LM. In order to train a deep bidirectional representation, we simply mask some percentage of the input tokens at random, and then predict those masked tokens. We refer to this procedure as a 'masked LM' (MLM). In all of our experiments, we mask 15% of all WordPiece tokens in each sequence at random.

Task #2: Next Sentence Prediction (NSP). Many important downstream tasks such as Question Answering and Natural Language Inference are based on understanding the relationship between two sentences, which is not directly captured by language modeling. In order to train a model that understands sentence relationships, we pre-train for a binarized next sentence prediction task.

3.2 Fine-tuning BERT. Fine-tuning is straightforward since the self-attention mechanism in the Transformer allows BERT to model many downstream tasks by simply swapping out the appropriate inputs and outputs. For each task, we simply plug in the task-specific inputs and outputs into BERT and fine-tune all the parameters end-to-end.

4. Experiments

We present BERT fine-tuning results on 11 NLP tasks. GLUE (General Language Understanding Evaluation) is a collection of diverse natural language understanding tasks. BERT_LARGE obtains a GLUE score of 80.5%, a 7.7% absolute improvement over the prior state of the art. On SQuAD v1.1, BERT achieves an F1 score of 93.2, outperforming the previous best system.

5. Ablation Studies

We ablate the effect of the number of training steps, and show that BERT does indeed require a large amount of pre-training to achieve high fine-tuning accuracy. We also compare bidirectional vs. left-to-right pre-training, showing that removing the masked LM's bidirectionality significantly hurts performance on tasks requiring context from both directions, such as SQuAD.

6. Conclusion

Recent empirical improvements due to transfer learning with language models demonstrate that rich, unsupervised pre-training is an integral part of many language understanding systems. Our major contribution is further generalizing these findings to deep bidirectional architectures, allowing the same pre-trained model to successfully tackle a broad set of NLP tasks.